Theoretical Perspectives on Social Relationships in Informal Financing

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ABSTRACT
Informal financing is commonly based on social relationships. We organize the theory of informal financing into three approaches that conceptualize those relationships as (1) local information, (2) social collateral, and (3) social preferences, respectively. All three are consistent with broadly observed patterns of informal financing. The social-preference approach can further explain certain distinctive aspects of financing from close family and friends. The various approaches reflect variety in informal finance, or better, the social mechanisms that enable it. We advocate for reframing the research agenda to move away from the dichotomy of informal-versus-formal finance and focus instead on how institutions in well-developed economies *combine* formal and informal mechanisms to unlock financing.

KEYWORDS
Informal financing, family financing, microfinance, crowdfunding

1. Introduction

When we think of where businesses obtain outside capital for investments, what usually comes to mind first are formal financial institutions or markets—like banks, stock and bond markets, venture capital, private equity—which exist for that purpose. However, in many emerging economies, a significant share of funding for small businesses comes from alternative, "informal" sources with borrowers tapping into personal relationships whose raison d'être often transcends the financing transactions.

The difference between formal and informal financing is that, in terms of design and implementation, the former relies on legal and contractual mechanisms, while the latter (further) leverages mechanisms based on other social interactions. This chapter reviews theoretical approaches to modeling the role of social relationships in business financing. The goal is to outline how our understanding of the peculiarities of informal finance has evolved.

We will begin by discussing a small (far from exhaustive) selection of recent empirical

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studies as a means to highlight some of the principal questions that motivate theories of informal business financing. We then review how the existing theoretical literature has conceptualized the role(s) social relationships could play in financing. We conclude our review with a synopsis and a suggestion for future research.

2. Empirical patterns and related questions

Beck, Demirgüç-Kunt, and Maksimovic (2008) analyze the financing sources of almost 3,000 firms in 48 countries using data from the World Bank Environment Survey. They find that informal financing accounts, on average, for only a small percentage of firms' investments, though it is more important for smaller firms and in weaker institutional environments, and exhibits substantial variation across countries. Overall, the patterns suggest that informal finance has some role to play, especially for small firms and when institutions are weak, but also that formal finance—and hence having institutions that facilitate it—is conducive and perhaps critical to large-scale business growth.

Ayyagari, Demirgüç-Kunt, and Maksimovic (2010) expand on this line of thought by studying a country where informal finance is seen to play a large role: China. They find that, while a large share of the 2,400 firms in their sample do rely on informal financing, the fastest-growing firms are the small percentage that (also) use bank financing. This is not entirely consistent with the influential thesis of Allen, Qian, and Qian (2005) that informal finance has been a catalyst for China's business sector despite a lack of formal finance.

The empirical patterns conform with the law-finance-growth paradigm that development of a formal financial sector is key to private-sector growth. But this does not imply that informal finance is unimportant; the patterns suggest that it may yet be crucial to the creation, sustenance, and growth of small firms—and most large ones started small. Revisiting the context of China, Degryse, Lu, and Ongena (2016) trace richer empirical patterns that hint at such dynamic nuances. They find that use of informal financing is associated with higher growth for small firms but lower growth for large firms. Further, combined informal and formal financing is associated with particularly high growth for small firms, but not for large firms.

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Allen, Qian, and Xie (2019) uncover further complexity in the patterns by separating underground financing by informal moneylenders from "constructive" informal finance, which includes trade credit and financing from family and friends. They show that firm growth *is* positively associated with informal finance—but only the constructive kind—thus reconciling the findings of Ayyagari, Demirgüç-Kunt, and Maksimovic (2010) with Allen, Qian, and Qian (2005). They also reaffirm that informal finance is more common for smaller, less established firms, and that informal and bank finance show patterns of complementarity for smaller firms (only).

Underscoring the important role of constructive informal financing in China is Allen, Qian, and Xie (2019)'s survey estimate that 67% of the financing at the startup stage is sourced from family and friends. It is instructive to compare this with startup finance in an economy with a more developed formal financial system. Robb and Robinson (2014) use data from the Kauffman Firm Survey to analyze the financing sources of 3,972 U.S. firms in their first year of operation. The main insight is that "funding from formal debt dwarfs funding from friends and family" (p.154) in terms of both total amounts and the number of firms that utilize the respective sources.[1]

These empirical patterns echo a number of central themes found also in earlier work.

P1. **Pecking order and firm life cycle**. Informal finance seems to play a vital role for smaller and younger firms, but becomes less prevalent as firms grow.

P2. **Co-financing**. For some firms, combining formal and informal finance seems to be optimal and particularly effective in supporting growth.

P3. **Financial deepening**. Informal finance appears to become less prevalent when the formal financial system becomes more developed.

P4. **Variety**. Different kinds of informal finance seem to vary in their effect on firms or when firms use them.

The empirical research cited above is ultimately interested in macroeconomic effects, namely the impact of formal and informal finance on (private-sector) economic growth. But patterns **P1**-**P4** hint at trade-offs in choosing between informal and formal finance

[1]The "average amount of bank financing is seven times greater than the average amount of insider-financed debt" (p.154). And in their sample of 3,972 firms, there were (only) 177 counts of spousal or parental equity, 327 family loans to main owners, 29 family loans to other owners, and 115 family loans to the business (Table 4).

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that require microeconomic theories to analyze. Micro-founding these trade-offs would frame how to interpret the link between (types of) finance and economic growth.

That said, the earliest microeconomic theories of informal financing—and the role of social ties in it—were more directly motivated by specific institutions, such as *rotating savings and credit associations* and *group-lending programs*, which were able to sustain financial arrangements in settings where formal financial institutions were less effective. Such institutions immediately raise questions about what makes them effective but also why they might nonetheless be insufficient to catalyze the kind of economic growth seen in modern economies.[2]

Some of the questions that theories of informal financing shed light on are:

- What are the comparative advantages and limitations of the respective sources of financing? In what way or in which circumstances are informal and formal sources and mechanisms substitutes or complements?
- Why do the various types of finance differ in their empirical associations with firm growth? How does this relate to firms' life cycles? Do the differences reflect causal effects or selection?
- With financial development, would formal finance eclipse or even reduce informal finance? If so, is obsolescence or scarcity driving the latter's declining share?
- What are the implications for public policies aimed at improving access to finance as a means to promote economic growth?

The next section organizes and reviews existing theories of informal financing with a focus on how the underlying social relationships are conceptualized; how they speak to the above questions; and how well they capture the traits of the most common source of informal finance: family and friends.

3. What "social mechanisms" underlie informal financing?

In any theoretical framework in which formal or informal finance can exist, or co-exist, each must have advantages *and* disadvantages vis-à-vis the other; otherwise, one would

[2]Besley, Coate, and Loury (1993, fn.2) cite earlier anthropological and economic studies of rotating savings and credit associations, and Murdoch (1999) provides an overview of early debates on (the promise) of microfinance.

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strictly dominate. Since mainstream finance theories are focused on formal institutions, the crux of any theory of informal financing is the question: What benefit *and* cost does informal finance have relative to formal finance?

In organizing the literature around this rather general question, we do not do justice to other interesting, more specific questions that many of the surveyed papers address. Instead we simply group all papers according to the approach they take to modeling the role social relationships play in financing. We will discuss implications, strengths, and caveats of each approach.[3]

3.1. *The "information" approach*

Basic trade-off. As standard corporate finance theory builds on models of imperfect and incomplete information, it is natural that the "first" generation of informal finance theories makes the assumption that informal lenders have an informational advantage. The premise is that social interactions breed familiarity or allow lenders to more closely monitor borrowers' activities. As a result, informal finance is hampered less by adverse selection or moral hazard problems than formal finance.

The disadvantage of informal finance in this literature is that it is costlier—be it due to monitoring costs or a higher opportunity cost of capital—or simply in limited supply. In some theories, the agents that share social relationships do not lend to each other, in which case the supply of informal financing is effectively zero, but a formal lender seeks to harness the information within those relationships through a group-lending contract. The common idea across these settings is that social relationships mitigate information frictions and thus reduce endogenous financing constraints, but that a financial market limited to transactions between socially connected agents severely reduces the potential pool, or sources, of capital available to borrowers.

Information benefits are the focus of the earliest papers on the potential role of social ties in financing (Stiglitz 1990; Varian 1990), and there have been many variations and

[3] An alternative strategy could be to remain agnostic about the precise mechanism that generates the trade-off by modeling the benefits and costs as reduced-form parameters. This would generate a framework that subsumes various types of informal-financing mechanisms. The downside is that some nuanced implications are potentially lost in this catch-all model. The upside is that a comparative analysis between formal financing and (all kinds of) informal financing can be made without restricting oneself to a specific mechanism. See Allen, Qian, and Xie (2022) for such a holistic approach.

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extensions of this approach since then (Banerjee, Besley, and Guinnane 1994; Prescott 1997; Jain 1999; Ghatak 2000; Ghatak and Guinnane 1999; Laffont 2003; Rai and Sjöström 2004; Giné 2011; Cason, Gangadharan, and Maitra 2012; Madestam 2014, among others).

Strengths. The above trade-off could explain **P1**, that is, why smaller, younger firms in less developed financial systems are more reliant on informal finance, if one assumes that the financing of such firms is subject to greater information frictions; or conversely, why larger, older firms tend to transition to formal finance as they need a larger supply of capital or based on the assumption that a formal lender's information disadvantage is smaller for such firms (e.g., as they have longer track records or an increased availability of collateralizable assets).

It could also explain **P2**. Firms would find it optimal to combine informal and formal finance; the former mitigates information problems and the latter offers complementary cheaper funding. Such complementarity could also be exploited through co-signed loans or group-lending schemes to harness the information socially connected borrowers have about each other (more on this in our discussion of microfinance below).

Last, if the formal financial sector grows and intermediates more capital, the relative share of capital raised through informal finance could drop. Moreover, if the growth of the formal sector comes with, or is driven by, improvements in its capacity to overcome information problems, the absolute volume of informal finance could shrink as well. In other words, this framework could also explain **P3**.

An appealing aspect of this theoretical framework is that it blends in seamlessly with other corporate finance theories where differences in information or monitoring abilities are the main distinction between alternative types of financing—such as, e.g., between bank finance and market finance (e.g., Holmstrom and Tirole 1997) or trade credit and bank finance (e.g., Burkart and Ellingsen 2004). The general paradigm underlying this literature is that variety in financing sources (or intermediaries) stems from variety in the levels or types of information that different financiers (or intermediaries) possess.

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Caveats. While consistent with the empirical patterns **P1-P3**, this framework has a few features that do not map well onto financing from family and friends, which is one of the most common sources of informal financing. The presence of monitoring costs or higher opportunity costs of capital (i.e., capital scarcity) implies that, at least for firms with both types of finance, informal financing should be more expensive (e.g., a higher interest rate).[4]

In many financial transactions with family and friends, however, interest rates as well as expectations of being (fully) repaid are lower than for financing from formal financial institutions. It is in this regard instructive to broaden the perspective and look beyond financing for business needs; in networks with strong social ties, we observe a variety of financial flows—such as, e.g., remittances, informal insurance, gifts, early bequests—that are, or at least appear, charitable. In fact, there are large literatures on those kinds of financial transfers "among family and friends" as well.[5]

The "information approach" to explaining informal business financing, if applied to family and friends, has no natural link to these other—equally, if not more, prevalent—types of intrafamily transfers. As much as it cannot explain "below-market" pricing of family loans, it also cannot account for outright gifts.[6] The ideal framework would have the scope to synthesize the various intrafamily transfers.

Last but not least, whether the premise of the "information approach" holds in all or even most business financing transactions is far from obvious. While it is arguably true that family or close friends may be better judges of a borrower's character, it is unclear whether they are—or this alone makes them—better at evaluating the prospects of the business to be financed. Just to picture a concrete context, is it plausible that when Jeff Bezos left a lucrative banking career (with a successful track record in various corporate settings) to start his internet venture in the early 1990s, his parents who provided about $250,000 in start-up funding were able to evaluate his endeavor better than professional investors? Do we believe that having superior information is why they financed Amazon at the time, when others did not? Or might they have done so for other reasons, in spite

[4]This is analogous to bank finance being pricier than market finance in, e.g., Holmstrom and Tirole (1997).

[5]To cite but a few works from said literatures, see Rapoport and Docquier (2006) on remittances; Udry (1994) and Fafchamps and Lund (2003) on informal insurance; and Kotlikoff (1988) on intergenerational transfers.

[6]Incidentally, the issue that a below-market or interest-free family loan is akin to a gift has a legal and judicial history in U.S. tax law (Hutton and Tucker 1985).

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of *not* being privy to more optimistic information about the venture?[7]

Microfinance. An important practical implication of the "information approach" is that formal lenders could harness the private information embedded in social networks through co-financing or co-signing arrangements whereby a borrower's family or friends provide part of the funding or assume part of the liability. Perhaps the most ambitious arrangements of this type are group-lending programs in which group members' joint liability gives them incentives to leverage information they have about each other, be it for mutual "screening" or "monitoring."[8]

Part of the promise of the microfinance movement was that formal lenders could use such group-lending contracts to harness the mechanism of informal finance without the latter's capital supply limitations. By scaling the benefit of social ties for financing, this could significantly relax financing constraints and promote business growth. However, a recent series of rigorous experimental evaluations of microfinance shows only a modest effect on entrepreneurial growth (Crépon et al. 2015; Tarozzi, Desai, and Johnson 2015; Banerjee et al. 2015, 2017, 2019). While a variety of factors could explain this finding, one interesting possibility is that certain aspects of involving social relationships (other than information benefits) in financing a business may pose obstacles to the borrower's risk-taking and growth aspirations.

3.2. *The "collateral" approach*

Basic trade-off. The second approach also borrows its central idea from mainstream finance theory: borrowers can mitigate incentive or information problems by involving collateral they lose in the event of default. Except in the context of informal financing, the loss can be a deterioration of social relationships; the relationships, in other words, constitute "social collateral."

The disutility the parties experience in the case of default or breach of trust can take

[7]The following passage from *The Everything Store* is telling in this sense (Stone 2013, Ch. 2):

> "We saw the business plan, but all of that went over our heads to a large extent," says Mike Bezos. "As corny as it sounds, we were betting on Jeff." Bezos told his parents there was a 70 percent chance they could lose it all. "I want you to know what the risks are, because I still want to come home for Thanksgiving if this doesn't work," he said.

[8]In fact, many of the papers cited earlier in this section are theoretical analyses of group lending schemes.

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various forms, manifesting itself in both emotions (e.g., shame, anger, disappointment) and actions (e.g., shunning, ostracizing, violence). The benefit of informal financing is that the risk or threat of such consequences improves the borrower's incentives ex ante, deterring misbehavior or dishonesty. The downside is, of course, that a default can lead to deadweight losses ex post if the relationship between lender and borrower sours.

Having said that, we should note that the limit of informal finance emphasized in the "information approach" applies equally in this theory, and for that matter, in any other where financing relies on an entrepreneur's social ties: the potentially accessible pool of investors is limited to those ties. While this is at least implicitly true in all models, there are notable network-theoretic models within the "social collateral" approach where this limitation is explicitly captured.

The pioneering theories emphasizing (social) exclusion, peer pressure, and sanctions in informal financial arrangements focused on rotating savings and credit associations (Besley, Coate, and Loury 1993, 1994) or group loans (Besley and Coate 1995; Abbink, Irlenbusch, and Renner 2006; Chowdhury 2007; Bond and Rai 2008; de Quidt, Fetzer, and Ghatak 2016). But the idea has also been applied to informal finance more broadly (Karlan et al. 2009; Karaivanov and Kessler 2018).

Note that models of the "collateral approach" appeared not much later than those of the "information approach." In fact, the approaches overlap significantly; in quite a few versions of the former, social sanctions require monitoring of outcomes (i.e., costly state verification), and in some of the latter, the narrative of monitoring appears to implicitly involve sanctions or pressure. This is perhaps unsurprising given that many pioneering papers were inspired by the success of the Grameen Bank and the low delinquency rates of its group-lending schemes, and descriptions of those schemes emphasized the role of information and social pressure within the groups (e.g., Hossain 1988).

Strengths. This framework implies that younger, smaller firms rely more on informal finance if one assumes that they have fewer pecuniary assets that can serve as collateral, and that more established firms with more collateralizable assets have greater access to formal finance. Also, given social collateral may not scale up with firm size, there may be a limit to the magnitude of funding that can be sustained by this type of collateral.

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As social collateral mitigates incentive problems through the risk of social sanctions, borrowers may only want to use as little of it as required and complement it with formal finance. Relatedly, a formal lender can harness the incentive benefits of social collateral through group lending mechanisms, a practical implication prominent in the context of microfinance (like the analogous implication of the "information approach").

Last, if social collateral is limited or does not scale with firm size, an expansion of the formal financial sector will lower the share of funding from informal sources, and if that expansion is driven by an improved capacity of formal institutions to address incentive or information problems through means other than social collateral, informal financing may even shrink in absolute terms. So, like the "information approach," the "collateral approach" is compatible with the broad empirical patterns **P1-P3**.

A significant feature of the "collateral approach"—one that distinguishes it from the "information approach"—is that not only the benefit but also the *cost of using informal finance originates from social interactions*. This is not about the parsimony of a theory; the deeper point is that involving social relationships can entail negative repercussions. For example, some reviews of group lending note that the aftermath of group members' default can be harsh, with social sanctions ranging from ostracization to violence.[9] The threat of such sanctions, or in the abstract, of losing social capital upon default could in turn discourage entrepreneurial risk-taking.[10]

This adds a new angle on the association of firm growth with various types of finance. Under the "information approach," informal finance has only benefits; any dissociation with growth stems from its limited supply. Cast in terms of a hypothetical experiment, randomly replacing formal funding partially with informal funding (and its information benefits) would not reduce the borrower's incentive to take efficient risks. By contrast, under the "collateral approach," an exogenous shift to informal funding (and its risk of sanctions) could have a negative *causal* impact on risk-taking. This calls into question whether involving social relationships in financing, even when possible, is always good.

Caveats. Unless the threat of sanctions is never carried out on the equilibrium path, informal finance should be more expensive than formal finance when both are used. The

[9]See, e.g., Montgomery et al. (1996, p.154f) or Ghatak and Guinanne (1999, p.221).
[10]See, e.g., Lee and Persson (2016, Sec.3) and Karavainov and Kessler (2018).

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reason is that the damage to a social relationship harms both sides.[11] That is, not only the borrower but also the lender perceives this risk as costly and thus prices it into the loan like any other expected loan loss. This is different from asset collateral, which leads to a positive transfer to the lender in the event of default and hence reduces the nominal interest rate she charges ex ante.

Thus, if applied to financing from family and friends, the basic "collateral" approach does not predict below-market loan pricing, nor can it readily integrate other common charitable transfers such as gifts, bequests, and remittances among family members. To account for such phenomena, the approach would have to be amended. For example, it might be instructive to also consider social consequences, or norms, that are triggered if *no* financial "help" is provided (rather than only those that are relevant after a default). In Section 3.3.2, we highlight another possible extension of the "collateral approach"—with non-monetary favors and explicit other-regarding preferences—that is compatible with below-market loan pricing.

Another limitation of the "collateral approach" is that it requires social relationships to be vulnerable, or lenders (group borrowers) to be inclined to impose social sanctions on defaulting borrowers (group members). However, to which extent social interactions are of such nature may vary across different cultural contexts. For example, Ghatak and Guinnane (1999, p. 220f) recount experiences of group-lending programs in areas where the degree of social cohesion made borrowers unwilling to pressure, let alone sanction, each other.

Even within a cultural context, the strength of social ties is not uniform. Familial ties are often stronger than other social ties, and the strength of ties varies within extended families as well. The "collateral approach" suggests that those social relationships *most* immune to strife over money issues are among the *least* suitable for supporting informal finance. Prima facie, it is far from clear, if not doubtful, whether this is true in reality.[12] Casual observation suggests that financing from family and friends tends to come from those closest to the borrower—and connecting this point to the first caveat, rather often

[11] This notion is reflected in the following quote from *Hamlet, Act I, Scene 3*, "Neither a borrower nor a lender be; for loan oft loses both itself and friend."

[12] Again using Jeff Bezos's example to illustrate this point, one wonders how much the relationship to his parents would have suffered if their investment in Amazon had not paid off, or to paraphrase his words, whether he would still be welcome on Thanksgiving (cf. footnote 7).

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with generous financial terms.

3.3. *The "social preferences" approach*

The first two approaches adapt central ideas from standard corporate finance theory—monitoring, information, collateral—to intuit what role social ties could play in finance. The next approach takes its cue from elsewhere, not finance but fields primarily focused on social relationships: family economics and the subfield of behavioral economics that studies prosocial behavior.

Motivated by longstanding theories of kin altruism in biology and social psychology, family economics has a long tradition of modeling familial and kinship ties as involving altruistic preferences, since at least Becker (1993). Naturally, directed altruism could be an explanation for gifts, bequests, and insurance—and many other acts of generosity or charity—observed within families.[13] It seems only logical to broaden this perspective to intrafamily financing as well.

Shifting the perspective in this direction is also consistent with experimental studies in behavioral economics on the determinants of prosocial behavior such as generosity in (monetary) transfers. For example, Leider et al. (2009) conduct experiments to identify motivations for prosocial behavior *in social networks*, and in particular, to differentiate "selfless altruism" and "expectations of reciprocal favors in future exchanges" (p.1816). They find that the subjects, on average, give their *friends*

- 52 percent more due to directed altruism alone, and
- an additional 24 percent more when there is a chance for reciprocation

than what they give to random strangers. So, both motivations—directed altruism and expected reciprocity—turn out to be important for prosocial actions in social networks, i.e., along social ties.

The third theoretical approach to informal finance is to model such social preferences to characterize social relationships. We will discuss two specific models from Lee and Persson (2016). Sections 3.3.1 and 3.3.2 will outline the key effects and the basic trade-

[13]This is not to claim that *all* familial ties or family situations are best characterized by such preferences, but rather than such preferences—when present—may be instrumental for intrafamily financing. For a critique that the characterization of familial relationships in the economics literature is sometimes too narrow or "idyllic," see, e.g., Pollak (2003, Sec.X).

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off in those models. Section 3.3.3 will discuss some of the implications that emerge from this approach.

3.3.1. Directed altruism

A theory of directed altruism within, for example, a family generates financial transfers even in the absence of a business need or investment opportunity. Such altruism creates intrafamily *insurance*; family members will protect one another against negative shocks to consumption.

This charitable attitude remains relevant when one introduces demand for financing into this framework; it translates into *below-market financing* terms if an entrepreneur depends on family funding for a risky investment or consumption-smoothing over time. In fact, small amounts may be contributed as outright *gifts*.

While a family investor's altruism towards the entrepreneur leads to cheap financing, the latter's altruism toward the former mitigates frictions that might otherwise hamper a financing arrangement: the entrepreneur is less inclined to misuse or divert resources, or mislead the investor. Hence, the family investor can passively *trust* the entrepreneur to abstain from misbehavior that, absent altruism, may require active countermeasures like monitoring, screening, or a threat of sanctions. Strikingly, if frictions are overcome by *trust*, family may be (the only ones) willing to back the entrepreneur even when they understand the venture *less* than formal investors do.

At first sight, informal finance seems to have no inherent disadvantage in this theory. But that is not the case. Although family financing is cheap and alleviates contracting frictions, the entrepreneur prefers formal financing and approaches family only if, or to the extent that, it is required by frictions. Likewise, when her access to formal financing improves, she would replace cheaper family financing with it.

The reason is that financing the risky venture reduces the family wealth available for intrafamily insurance. In essence, family finance replaces one kind of generous transfer with another—but while insurance reduces (individual) family members' consumption risk, sinking family funds into a risky venture does the opposite. Hence the (risk-averse) family would rather keep the informal insurance in place and use non-family financing,

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if available, for the venture.[14]

Thus, the basic trade-off is the following: On one hand, while family financing seems cheap in terms of (below-market) pricing, the *shadow cost* is that it risks funds available for (also charitable) intrafamily insurance. On the other hand, family relationships can capitalize on *trust* to overcome frictions that render formal financing infeasible.[15]

3.3.2. Social debt

For certain social relationships, a model with *unconditional* directed altruism is inapt. A plausible alternative may be that the altruistic sentiments, and derivative acts of help or generosity, rely on expectations of reciprocity and diminish if those are disappointed. Generous interactions among friends are then repeated "gift exchanges" through which they pay and repay each other favors of different kinds and in different "currencies," be it money, material gifts, time, emotional attention, or physical effort.

Lee and Persson (2016) consider a reduced-form version of such preferences in which violations of the gift exchange norm between friends reduce the altruism between them. The norm is violated if a favor is not repaid despite there being various ways of doing so. That said, not-in-kind or non-monetary repayments do involve deadweight costs in that they can be less effective or inconvenient means of repayment.[16]

In this model, borrowing from friends creates a financial liability that, in the event of default, becomes "social debt." This effectively eliminates limited liability and replaces it with a kind of "social peonage"[17] enforced by the potential damage to the friendships (i.e., social collateral). Financial default is hence costly to the borrower not only if the friendships are damaged but also if she opts to salvage them by paying the social debts; it puts a *strain* on the friendship or makes it *strenuous* to preserve.

This social-debt effect has several implications. First, a financial loan may be priced

[14]To be more precise, given the family's aggregate indirect utility function (incorporating contingent insurance transfers) is concave, investing family funds into a risky venture decreases the expected family utility relative to formal financing (from an outsider) which, if priced competitively, affords the family the net present value of the investment without reducing the family funds available for intrafamily insurance.

[15]If altruism is modeled in a "paternalistic" way (taking other's payoffs rather than their utilities into account), another shadow cost of family financing is that borrowers experience disutility from exposing family members to financial risk—a form of "social risk aversion" (Lee and Persson 2016, Internet Appendix).

[16]The "debtor" could also end up paying back more if the valuation of not-in-kind (or non-monetary) favors is more difficult to assess (in monetary terms).

[17]Peonage, also known as debt bondage, debt slavery, or bonded labour, is, loosely speaking, a system in which a debtor can be compelled to pay off the debt with work. When the service required to repay the debt is not clearly defined, there is sense in which the creditor has control over the debtor('s freedom).

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below market or at zero interest rate even if there is a non-zero probability of a financial default, as repayment may come in the form of other social (debt) interactions. So while this model also involves the notion of social collateral, as the approach in Section 3.2, it is compatible with below-market pricing of loans between friends.

Second, the absence of limited liability improves borrower incentives. This mitigates contracting frictions that might render financing outside of these friendships infeasible. A curious prediction of the social-debt model is that, unlike in the basic social-collateral model (or standard financing theory for that matter), the borrower's incentives *improve* when the loan is larger; the reason is that the larger social debt resulting from a default acts as a stronger deterrent.[18]

Third, even though a loan between friends may be priced below market and mitigates contracting frictions, formal financing is preferred when it is available. This is because repaying *social* debt—while beneficial for incentives—involves deadweight costs for the borrower. This *shadow cost* is larger (in expected terms) when financial default is more likely. Such financing is hence less appealing for riskier investments.

3.3.3. Distinctive implications

Both of the social-preference models are consistent with empirical patterns **P1** to **P3**. If younger or smaller firms face greater contracting frictions, they have to rely more on informal finance. Conversely, larger and more established firms would have more access to formal finance, which they prefer due to the shadow costs of informal finance. This inverse relation between firm size and informal finance is further reinforced by the effect that the shadow costs discourage entrepreneurial risk-taking (in the pursuit of growth). That is, informal finance may *causally* slow or inhibit firm growth—though without it, small firms may not grow or emerge at all.

Due to the shadow costs, entrepreneurs would limit informal financing to the amount needed to overcome frictions, and if possible, complement it with formal financing. This combination overcomes financial constraints with informal "trust" capital, while at the same time leveraging as much formal "risk" capital as possible. This financing mix may

[18]This effect holds up to a loan amount at which the borrower would rather forfeit the relationship than pay the social debt, in which case the social-debt model reverts back to the basic social-collateral model.

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be a particularly potent catalyst for growth in early stages of a firm's life cycle.

Last, since the shadow costs make informal finance generally less attractive (unless it is needed), its use may decrease as the formal financial sector becomes more developed. However, the social-preference models also suggest that (the use of) formal mechanisms could reduce the shadow costs of using informal finance and thus "unlock" *more* of it—a point we elaborate on further below.

So the social-preference models are also consistent with the broad empirical patterns. However, they exhibit a number of additional aspects that are noteworthy:

Unifying perspective. A nice feature of both models is that they pertain not only to informal financing but integrate other types of transfers within the social relationships. The directed-altruism model predicts a number of intrafamily transfers seen in practice (such as insurance, gifts, and below-market financing), and the social-debt model treats financing between friends as but one (type of) transfer within a repeated gift exchange, which involves all kinds of generous acts within the friendship.

Both the assumptions that create the unified perspective and the interdependence of various transfers are crucial to the main effects and trade-offs in those models. In the first model, altruism is the root of intrafamily insurance and below-market financing and the shadow cost of the latter is that it displaces the former. In the other model, the gift-exchange norm spans a variety of reciprocal transfers, and in the event of a financial default, leads to expectations that repayment be made through other means and (non-financial) favors—social debt that here constitutes the shadow cost of informal finance.

Price-demand paradox. Both models can account for the fact that financing from family and friends is commonly priced below market, but at the same time predict that borrowers do not prefer such financing when the intended use for those funds is risky. The models can thus explain why small business experts usually warn of the dangers of borrowing from family and friends, even when the terms are generous, or why budding entrepreneurs may hesitate to burden family and friends with the financial risk of their venture.

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Use of funds. The models predict that the shadow costs of financing from family and friends—and hence the borrower's reluctance to use it—decrease when the use for those funds is less risky. This implies that, empirically, family and friends should be the first resort when one is plainly in need of plain help or when looking to smooth consumption over time, but the last resort when one is looking to finance a risky venture. So, while the models predict that family and friends are not the best source of *risk* capital, they are consistent with evidence (1) that many approach family and friends first when they need a lump sum and (2) that family and friends are the predominant source of informal insurance.

Benefit of formal institutions. While much of the existing literature (following the previous approaches) focuses on the benefits of harnessing social ties for financing, the social-preference models shifts the focus to the benefits of formal financial institutions. Specifically, the models suggest that the development of formal institutions can reduce the shadow costs of informal financing and thereby make the latter *more* attractive.

For example, in the social-debt model, the use of formal third parties to collect debt can reduce social tensions that undermine the inclusion of large loans in gift-exchange relationships that rely only on "handshakes" (Lee and Persson 2016). And the directed-altruism model suggests that, if the development of formal insurance decreases the need for informal insurance, family funds become a more suitable source of risk capital.

The general point is that the development and involvement of formal institutions can (1) alter the scope of financial transfers within families and (2) moderate tensions that arise from entangling social relationships with financial arrangements. Both effects can potentially unlock *more* family financing for risky ventures—in absolute terms—as the formal financial sector becomes more developed.[19] They also raise the question to what degree the ways in which, or the channels through which, social networks are leveraged for business financing evolve with economic development.

[19] In *relative* terms, informal finance might still grow less important if formal institutions improve.

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4. Concluding remarks

4.1. *Synopsis of the literature*

In our reading, the literature on informal financing is bifurcated into two strands—one with a "macro" and the other with a "micro" orientation—both of which are interested in questions of practical significance. Emerging from the law-finance-growth paradigm, empirical work on the relative significance of informal and formal financing is interested in how critical the development of the formal financial sector is for economic growth, or conversely, to what extent economies can exhibit substantial growth if the formal sector is underdeveloped and financing is primarily informal.

A parallel literature in development economics delves into specific types of informal finance (e.g., rotating savings and credit associations or group lending) with a related, but more instrumental, focus on *how* informal finance utilizes social ties to be feasible in environments where (purely) formal finance is not. Given this "micro" focus, it is this literature that has produced most of the theoretical work on social ties and finance.

Both theoretical arguments and the empirical evidence suggest that the development of formal financial institutions is a catalyst for economic growth. One argument is that informal finance dampens entrepreneurial risk-taking but the strongest argument is the simplest: financing that relies on social ties is limited by the social network structure.[20] A formal sector that facilitates arm's-length transactions can transform an economy in which (only) "friends finance friends" into one in which "strangers finance strangers." This does not rule out that a society with dense networks—i.e., strong social capital—can generate substantial economic growth. Yet, even in such contexts, the ability to tap into a pool of investors beyond one's own social circle unlocks further growth potential for firms that require large amounts of capital.[21]

At the same time, the theories imply that informal financing may yet be crucial when contracting problems are most severe, which is in consistent the empirical pattern that

[20] This is explicitly captured in the network-theoretic models within the "collateral" approach. In early models, particularly within the information approach, this logic is captured in reduced form in the assumption of limited capital supply or higher cost of capital on the part of informal financiers.

[21] The above argumentation echoes the central themes in the literature on informal insurance networks which, on one hand, ponders why these networks achieve a surprisingly high degree of risk-sharing and, on the other hand, ponders why they are not fully efficient. Ambrus, Mobius, and Szeidl (2014) and Ambrus, Gao, and Milan (2022) are examples that illustrate this balance of perspective.

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informal financing seems more important for young and small firms. That is, while none of the reviewed theories explicitly feature a dynamic life-cycle model of firm financing, there is a strong connotation that informal finance—or the underlying social ties—can be crucial at the outset of some ventures, even if later-stage growth of those that prove successful may depend on their ability to tap into a greater pool of capital through the formal sector.

The theories identify several possible mechanisms through which social relationships may lubricate financing. We see this less as theoretical ambiguity than as a reflection of the fact that social relationships are multi-faceted such that different aspects, or types, of relationships can be relevant. From a meta-perspective, the different approaches may in fact portray *variety* in informal finance that exists in practice (Hulme, Montgomery, and Bhattacharya 1996, p.200):

> Sanderatne and Senanayake (1989) identify a wide range of intermediaries in the informal market, including professional moneylenders, pawnshop operators, friends and relatives, traders, commission agents, cheetus (a form of rotating savings and credit association) and special-purpose credit groups.
>
> Given the differing motivations, objectives and modus operandi of such intermediaries the informal market is characterised by widely ranging explicit and implicit interest rates, as well as widely varying terms and conditions (Zander 1992). Crude estimates of the charges levied by informal lenders suggest that around one-third of loans are interest-free (that is, a negative real rate), around one-third are between 1 and 25 per cent per annum (that is, charging below and around a real rate) and one-third are above 25 per cent per annum. For a proportion of the latter, rates are very high running into hundreds of per cent.

The micro-theoretic focus on the different roles social ties can play further suggests that they may remain partly relevant even in developed financial systems. *Information* embedded in, or exchanged through, social relationships will remain important. So will both *altruistic sentiment* between people close to one another and *informal governance mechanisms* in social networks, such as *reciprocity norms* and *social sanctions*. It would be surprising not to see family or friends, and other kinds of social relationships, being a key source of financing in at least some contexts and situations, even in quite developed

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economies (like, e.g., the United States). Notions that formal finance (not just outgrows but) renders financing based on social ties irrelevant may be an unintended connotation of the macro-lens of the law-finance-growth paradigm, which in turn may obscure more nuanced questions that arise from the variety of micro-theories.

4.2. *Evolving institutions and role of social ties in finance*

The research in this literature seems to be framed in terms of two key questions: How much can informal finance support economic growth, and how much importance does it have or retain when the formal financial sector becomes more developed? We think that this language imposes conceptual restrictions that may block a deeper understanding of the subject, and that further progress could come from reframing the subject along two dimensions to move way from the dichotomy of "informal" versus "formal" financing:

(1) **Variety is key**. Informal finance is not a monolithic phenomenon and the reason is that social relationships can affect financial contracting through a diverse set of mechanisms. Considering the various mechanisms, or roles of social relationships, rather than "informal financing" *per se* opens up a richer perspective.

(2) **Focus on mechanisms.** With an emphasis on mechanisms, a dichotomous view of informal versus formal finance can be replaced by a integrated view of informal and formal mechanisms. The focus shifts from asking what type of finance is more important to whether and how mechanisms can be optimally combined to enable financing.

The perspective laid out by (1) and (2) may seem intuitive but it is somewhat lost when a micro-theory magnifies a single aspect of social relationships or the macro-view lumps together different kinds of informal finance.

One application of this perspective is to transform the question whether the informal financial sector diminishes as the formal financial sector develops into whether and how the role of social relationships in finance evolves with formal mechanisms that become available. This angle is particularly salient in light of the micro-theoretical implications regarding the benefits of combining formal and informal mechanisms:

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- Several papers in the information and collateral approaches study intermediaries that harness informal social ties when lending out formally sourced capital.
- Both models in the social-preference approach suggest that combining formal and informal finance relaxes constraints while promoting entrepreneurial risk-taking.
- The directed-altruism model suggests that the development of formal insurance markets may unlock family funds for business financing.
- In the social-debt model formal intermediation reduces social tensions that would undermine purely informal financing between socially-connected parties.

Let us discuss one historical and three more current examples to illustrate that the link between "formal" and "informal" finance could be more complex than the dichotomous terminology tends to suggest.

First, some formal financial institutions have historical roots in informal ones. Many savings and loan associations in the United States seem to have emerged from rotating savings and credit associations.[22] In fact, the same holds true for many credit unions in other parts of the world. One notable example are the rural credit unions in Germany founded in the 1800s, e.g., by Friedrich Wilhelm Raiffeisen. These cooperatives strongly relied on the social relationships—referred to as the "bonds of association"—among its members. That is, financial contracts were enforced by the social ties in the community, as colorfully explained by one historian (Wolff 1910, p.29f):

> It is quite true, as Dr. Johnson unkindly reminded Goldsmith, that it takes 240 poor men's pence to make one capitalist's sovereign. But once the sovereign is so put together, it is a totally different sovereign from that taken out of the rich man's purse. It has behind it 240 wills, 240 pairs of watchful eyes, 240 thinking brains. It has, so to speak, become an animate sovereign, with prudence, energy, vigilance, diffused throughout all its parts. Every spring, every wire of the composite machine takes a personal interest in the collective doings, watching the other parts, guarding against loss and waste, correcting the slightest irregularity.

In effect, this meant that delinquent borrowers faced social sanctions, be it shaming or even ostracizing.[23] While such severe sanctions may be less common today, many credit

[22] Besley, Coate, and Loury (1993) note this with references to Symons and White (1984) and Grossman (1992).
[23] See, e.g., Guinnane (2001).

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unions still operate in small communities and their core mission remains to improve the welfare of their local community rather than maximize profit. Both the potential costs and benefits of such cooperatives—in finance or elsewhere—hinge on the amalgamation of formal organization and informal relationships.

Second, crowdfunding platforms are modern formal financial institutions, but it has been shown that they harness social relationships and so combine formal and informal mechanisms. Early investors on crowdfunding platforms are usually family and friends (Agrawal and Goldfarb 2011), and more broadly, community participation seems to be an important aspect of crowdfunding.[24] In addition, there is a growing body of research that highlights how various aspects of social networks matter in crowdfunding, such as e.g., altruism, reciprocity, and local information.[25] On the other hand, it is interesting that crowdfunding platforms formally intermediate between parties, such as family and friends, that could in principle enter an informal financing agreement "off-platform." Some of the reviewed informal finance theories suggest that formalizing the contracts—e.g., with third-party auditing and debt enforcement—can mitigate the shadow costs of mingling business and friendships. In a way, formal intermediation helps depersonalize certain aspects of the transaction and so partly separates the financing of the enterprise from interacting with the entrepreneur to balance costs and benefits of involving social relationships.

Third, venture capital (VC) is a financial innovation developed in economies with a well-functioning formal financial system. Many aspects of complex VC contracts, with their dynamic and contingent allocations of cash flow and control rights, rely on strong formal institutions to be implementable. At the same time, it is well known that social networks play an important role in the VC ecosystem. Consider this short excerpt from *The Holloway Guide to Raising Venture Capital* (Sparks and Jepsen 2021):

> New entrants to Silicon Valley may be surprised to find that many people go out of their way to help each other—especially new founders—without expectation of getting anything in return. A well-known camaraderie exists between individuals who have made the radical decision to forgo a lot of the trimmings of a "normal" life; many founders try

[24]See Agrawal, Catalini, and Goldfarb (2014) for a survey article of crowdfunding.
[25]See, e.g., Vismara (2016), André et al. (2017), Giudici, Guerini, and Rossi-Lamastra (2018), and Polzin, Toxopeus, and Stam (2018).

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> to pay back the help they've gotten along the way by helping others. Venture capitalists also act as mentors to founders, and strategy meetings over beers are not uncommon. To many, the power of the network is one of the most welcome and unique characteristics of the startup world ...
>
> Networking as it relates to venture capital is particularly important for a few reasons. Once they've invested, an investor and a founder will build a relationship for years, sometimes a decade or more. Building trust and a relationship before an investment is made is part of the VC ethos ... Venture capital is a people business, and VCs take the networks they've built very seriously.

Academic research confirms that networks are an important determinant of VC success (Hochberg, Ljungqvist, and Lu 2007). It appears as if VC thrives on the combination of strong formal institutions and informal social networks. As VC expands into emerging economies with less developed formal sectors, its reliance on networks will increase but its overall effectiveness will likely decrease (Ahlstrom and Bruton 2006).

Last, we want to speculate on indirect links between informal and formal financing that arise because transfers from family and friends can take many forms. For example, parental financial help for (first-time) home purchases is not uncommon, especially if parental wealth is high, be it through contributions to down payments or by co-signing the mortgage. If a borrower receives such financial help for her home purchase but then uses formal financing to launch a business, it becomes an exercise in mental accounting to argue that the family has not contributed *entrepreneurial* financing. More generally, any financial assistance that a budding entrepreneur receives—even if it is not strictly speaking financing for the firm—can increase the latter's residual financing capacity. It is therefore, in principle, possible that "informal" business financing does not decline in developed financial systems even if *direct* informal funding for firms decreases. Instead, it could be that transfers from family and friends nominally serve different, perhaps less risky, purposes—yet raise an entrepreneur's financing capacity in the formal sector.

The overall hypothesis we are formulating with these examples is that, as the formal financial sector becomes more developed, roles played by social ties in informal finance begin to manifest themselves through other channels. Hence even if traditional informal financial institutions *per se* vanish, the underlying social relationships remain relevant

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for financing, but in new ways that combine informal with formal mechanisms. This is a possibility—hinted at by the existing theories—that is worth testing and tracing out in more detail and, in our opinion, offers a richer and deeper framework than the standard dichotomy of "informal" versus "formal" finance.

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